A Conditional Coupling–Diversity–Precision Bound for TTRL-Based LLM Evaluation

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Abstract

We study a conditional coupling–diversity–precision bound for a specified TTRL observation model. The primary theoretical estimand is the coefficient of variation of a regular estimator of a scalar coupling parameter γ under independent Bernoulli observations with logistic success probability. Under this model, the Cramer–Rao inequality yields a precision floor. A positive product bound additionally requires a compact parameter space and an assumed positive lower bound on strategy entropy; it is not a universal impossibility theorem for arbitrary evaluators or protocols. Empirically, all eleven condition rows are reconstructed from the persisted per-repetition archives under a single estimator (per-repetition mean γ; full-archive CV, with five-seed subsample ranges), so the consolidated paper supports complete γ/CV condition-level verification on the reconstruction. The observed grid contains no point in the region $\{\gamma < 0.2, \mathrm{CV} < 0.3\}$; this emptiness is robust to threshold perturbations under the full-archive CV estimator (both low-γ rows have full-archive CV ≥ 2.3); five-seed subsample CVs can be lower, so subset-level emptiness is estimator-dependent. Chatbot Arena correlations are observational external associations rather than validation of the TTRL bound. The contribution is therefore a checkable conditional model, an explicit theory-to-estimator map, and a list of robustness tests required before broader generalization.

Reproducibility and ethics. The analysis combines persisted API archives with deterministic reanalysis and records SHA-256 checksums for the frozen evidence used by the reported numerical checks. Public code and license metadata remain submission-time decisions and are not asserted here. This work studies evaluation dynamics in LLM systems; ethical implications include silent reliability decay in safety-critical systems and the need for version-dependent evaluation verification.

When large language models serve as evaluators in adaptive systems, their biases propagate into agent strategy distributions through Test-Time Reinforcement Learning (TTRL). This evaluator preference coupling creates a measured tradeoff: the source-study five-row table spans $\gamma \in [0.033, 1.038]$ and $\mathrm{CV}(N{=}5) \in [0.157, 2.42]$, while the archive-based reconstruction (Table 3) spans $\gamma \in [0.033, 1.176]$ and full-archive CV $\in [0.196, 5.477]$: stronger evaluator signal produces more stable measurements (Ablation max: $\gamma = 1.038$, $\mathrm{CV}(N{=}5) = 0.157$) but introduces more bias (lower strategy diversity); weaker evaluator signal preserves strategy diversity (DS Self-Eval: $H = 0.999$) but produces noisy, unreliable measurements ($\mathrm{CV}(N{=}5) = 2.42$, a $15.4\times$ inflation).

Scope note. The result is conditional on the stated logistic observation model. A universal impossibility claim for arbitrary evaluators or protocols would require assumptions that are not

established here, including a compact parameter space and a positive lower bound on strategy entropy.

We ask when a coupling–diversity–precision product has a positive lower bound. Proposition 3 answers this only for the stated logistic observation model and entropy-floor assumption. The empirical grid is used to illustrate that conditional result, not to establish it.

Our contributions are:

1. A conditional precision bound with all observation-model and regularity assumptions stated explicitly.
2. A theory-to-estimator map distinguishing measured quantities from modeling assumptions and empirical heuristics.
3. A scoped empirical audit of the five condition rows reproduced in this manuscript plus observational Chatbot Arena associations.
4. **A standardized protocol** for within-condition verification with convergence dynamics analysis.
5. A robustness agenda for alternative metrics, thresholds, binning, exclusions, and leave-one-condition-out analysis.

1 Related Work

Adjacent agent-side constraints. Our results concern the measurement side of multi-agent evaluation; three recent lines constrain complementary surfaces. Bounded-memory testbeds for long-horizon agents Cheng et al. (2026) formulate a memory contract for the agent side, mirroring how the empirical grid here bounds the measurement side; their interaction—whether an explicit memory contract helps populate the favorable regime—is unexplored. Self-evolving rubrics for skill use Zhu et al. (2026) and explicit agent abstention signals Luo et al. (2026) are two further axes that could shift the triangle's empirical location and warrant investigation. **This work.** We contribute the measurement-side grid itself: a positive product floor under explicit logistic assumptions, a reproduced negative γ–H association across the empirical grid, and no observed point in the favorable region $\{\gamma < 0.2,\ \mathrm{CV} < 0.3\}$.

2 Framework

2.1 The EPC Framework

An executor agent maintains a weight vector $\mathbf{w} \in \Delta^{K-1}$ over K strategies. At each round t:

1. Sample strategy $s_t \sim \mathbf{w}^{(t)}$
2. Generate response using strategy s_t
3. Receive evaluator judgment $y_t \in \{0, 1\}$
4. Update weights: $w_{s_t}^{(t+1)} = w_{s_t}^{(t)} \cdot \exp(\alpha_{y_t})$ (multiplicative reweighting)

2.2 Core Definitions

Definition 1 (Evaluator Coupling)**.** *The coupling coefficient*

$$\gamma_A \to B = \frac{\|\mathbf{w}_A \to B - \mathbf{w}_B\|_2}{\|\mathbf{w}_B\|_2}$$

measures how much evaluator A's preferences distort executor B's strategy distribution. $\gamma \approx 0$ indicates no influence; $\gamma > 1$ indicates influence exceeding natural strategy concentration.

Definition 2 (Strategy Entropy)**.** $H = -\frac{1}{\log K}\sum_{i=1}^{K} w_i \log w_i$ *is the normalized Shannon entropy of the strategy distribution, measuring diversity Cover and Thomas (2012). $H = 1$ indicates uniform distribution; $H = 0$ indicates deterministic strategy selection.*

Definition 3 (Measurement Noise)**.** $\mathrm{CV}(N) = std(\hat{\gamma}_N)/\mathbb{E}[\hat{\gamma}_N]$ *is the coefficient of variation of the coupling estimate at sample size N, measuring the stability of coupling measurements; std is the sample standard deviation (Bessel-corrected denominator).*

3 Conditional Observation Model and Bounds

3.1 Observation Model and Regularity Conditions

Let $Y_1, \ldots, Y_N$ be conditionally independent Bernoulli observations with

$$P_\gamma(Y_i = 1) = s(\gamma) = \{1 + \exp(-\gamma)\}^{-1}, \qquad \gamma \in \Theta = [\gamma_\min, \gamma_\max],$$

where $0 < \gamma_{\min} < \gamma_{\max} < \infty$. The sampling unit is one independent seed-level evaluator outcome; repeated rounds within a seed do not satisfy this independence assumption and must first be aggregated or modeled hierarchically. We assume an unbiased regular estimator $\hat{\gamma}_N$, differentiability of the likelihood, parameter-independent support, finite Fisher information, and interchangeability of differentiation and integration. These assumptions describe a stylized scalar logistic model, not the raw multi-strategy TTRL trace.

Proposition 1 (Conditional precision floor)**.** *Under the model and regularity conditions above,*

$$\mathrm{CV}(N) = \frac{\sqrt{\mathrm{Var}(\hat{\gamma}_N)}}{\gamma} \geq \frac{1}{\gamma\sqrt{N s(\gamma)(1 - s(\gamma))}}.$$

Proof. For one observation, $\ell_i(\gamma) = Y_i \log s(\gamma) + (1 - Y_i)\log(1 - s(\gamma))$. Since $s'(\gamma) = s(\gamma)(1 - s(\gamma))$,

$$\frac{\partial \ell_i}{\partial \gamma} = Y_i - s(\gamma), \quad I_1(\gamma) = \mathbb{E}_\gamma[(Y_i - s(\gamma))^2] = s(\gamma)(1 - s(\gamma)).$$

Conditional independence gives $I_N(\gamma) = N I_1(\gamma)$. The Cramer–Rao inequality Lehmann and Casella (1998) gives $\mathrm{Var}(\hat{\gamma}_N) \geq I_N(\gamma)^{-1}$; dividing its square root by $\gamma > 0$ yields the result. □

Proposition 2 (Variance of the empirical distance proxy)**.** *Let w_t follow the multiplicative update $\log w_{t+1,i} = \log w_{t,i} + Z_{t,i}$, where Z_t are independent across rounds with $\mathbb{E}[Z_i] = \mu_i$ and $\mathrm{Var}(Z_i) = \sigma_i^2$ (the Bernoulli preference draws of the TTRL protocol), and let $\hat{\gamma}_N = \|w_A - w_B\|_2/\|w_B\|_2$ be the empirical coupling proxy at horizon N with normalized weights. Then, to first order in the Delta method,*

$$\mathbb{E}[\hat{\gamma}_N] = g(\mu) + O(N^{-1}), \qquad \mathrm{Var}(\hat{\gamma}_N) = \frac{1}{N}\sum_j \left(\frac{\partial g}{\partial \mu_j}\right)^2 \sigma_j{}^2 + O(N^{-2}), \tag{1}$$

where g is the normalized-distance map and the first-order term scales the per-round preference variance by the sensitivity of the proxy to each strategy's drift. In the independent-domain simulation described below (K=11, win $\times 1.08$ / lose $\times 0.96$, $\mathbb{P}(win) = s(\gamma)$), the realized $\mathrm{Var}(\hat{\gamma}_N)$ grows approximately linearly in N ($0.00034 \to 0.0022$ as N:5→30), because the proxy distance itself grows

with the trajectory length and its per-round variance does not average out; meanwhile the Bernoulli Cramér–Rao floor $1/(Ns(\gamma)(1-s(\gamma)))$ *decays as* $1/N$*. Hence*

$$\frac{\mathrm{Var}(\hat{\gamma}_N)}{1/(Ns(\gamma)(1-s(\gamma)))} \sim N^2 \cdot c(\gamma), \qquad c(\gamma) \approx 1.4 \times 10^{-5} (= 1.4e-5)$$

(fitted from the simulation: $\mathrm{Var/CR} = 0.013$ *at* γ=1, N=30, *i.e.,* $0.013/30^2$*),* (2)

so the conditional CR floor is severely non-tight for the trajectory proxy: the bound is far below the actual variance and the gap widens with N*.*

Proof sketch (Delta method). Write $\log w_t = \log w_0 + \sum_{s=1}^{t} Z_s$; the CLT for the random walk gives $\log w_N = \log w_0 + N\mu + \sqrt{N}\mathcal{N}(0,\Sigma)$ asymptotically, where $\Sigma = \mathrm{diag}(\sigma_j^2)$ under independent strategies. After normalization, $\hat{\gamma}_N = g(\bar{\mu}_N)$ with $\bar{\mu}_N \to \mu$; a first-order Taylor expansion about μ yields the mean and variance expressions above. The N dependence of the realized variance is set by the proxy map g: when the distance grows with trajectory length (independent-domain training, simulated), the leading variance term carries an extra factor of N relative to a mean estimator, and the ratio to the $1/N$ Bernoulli floor grows as N^2. When instead the sampling units are the Bernoulli preferences themselves (e.g., a direct logistic estimate of γ), the standard CR analysis of Proposition 1 applies and is tight in the regular-estimator regime. The empirical E4 archive (n=30, $\mathrm{Var}(\hat{\gamma})$=0.158, floor 0.176, ratio 0.90) sits between these extremes, indicating that the live coupling process includes domain-transfer structure that the independent-domain simulation omits. □

Estimator definition and bias. The proxy $\hat{\gamma}_N = \|w_A - w_B\|_2 / \|w_B\|_2$ is a distance between normalized weight trajectories, not a logistic estimate of γ; it is therefore not an unbiased estimator of γ in general. In the independent-domain simulation, $\mathbb{E}[\hat{\gamma}_N]$ is governed by the trajectory length N (its realized range is ≈ 0.08–0.20 across $\gamma \in [0.05, 3.0]$ at $N \in \{5, 10, 30\}$) rather than by the true coupling strength, so the proxy carries almost no information about γ under independent training. Consequently the standard Cramér–Rao analysis of Proposition 1—which assumes an unbiased regular estimator of γ—does not apply to $\hat{\gamma}_N$ in that regime; Proposition 2 reports the variance of the proxy as an empirical quantity and shows that the Bernoulli floor is both non-tight and inapplicable to it. When the data-generating process includes domain transfer (as in the live E4 archive), the seed-level variance of the proxy approaches the Bernoulli floor (Var/floor ≈ 0.90 at n=30), indicating that the proxy becomes informative about γ and that the floor is approximately tight for the seed-level sampling units. Readers should therefore interpret Proposition 2 as a caveat about the proxy, not as a CR-efficiency claim for it.

Empirical coverage of the precision floor and MLE closeness. We probe the proposition with a deterministic simulation and with the eleven persisted archives (script and output provided in supplementary material). **Coverage:** simulating $Y \sim \mathrm{Bernoulli}(s(\gamma))$ and estimating $\hat{\gamma}$ by the logit of the sample proportion (5,000 runs per cell), the empirical CV fails to meet the lower bound $1/(\gamma\sqrt{Ns(\gamma)(1-s(\gamma))})$ in 10 of the 21 probed cells (48%): $(\gamma, N) = (0.05, 30)$ (6.794 vs. bound 7.305), $(0.1, 10)$ (6.015 vs. 6.333), $(0.25, 10)$ (2.422 vs. 2.550), $(1.0, 5)$ (1.007 vs. 1.009), $(1.0, 10)$ (0.683 vs. 0.713), $(2.0, 5)$ (0.426 vs. 0.690), $(2.0, 10)$ (0.326 vs. 0.488), $(3.0, 5)$ (0.238 vs. 0.701), $(3.0, 10)$ (0.180 vs. 0.496), and $(3.0, 30)$ (0.186 vs. 0.286); the remaining 11 cells meet the bound. The pattern is uniform: the logit estimator has first-order finite-sample bias $\mathbb{E}[\hat{\gamma}] - \gamma \approx (1-2s)/(2Ns(1-s))$, which inflates with $1/(Ns(1-s))$; failures concentrate where N is small (N=5–10 at $\gamma \geq 0.1$) or where $s(\gamma)$ approaches the boundary (γ=3, s=0.953, fails even

at N=30). The estimator is therefore not the unbiased regular estimator assumed by Cramer–Rao in exactly the cells where the bound matters most. The proposition is a statement about the regular-estimator regime, not a universal inequality—matching the conditional framing of the observation model above and of the product bound below. **MLE closeness:** fitting the Bernoulli model $Y_i = \mathbf{1}[\gamma_i > 0]$ to the eleven archives yields a finite logit-MLE in only two conditions (DS Self-Eval: $\hat{\gamma}_{\mathrm{MLE}} = -3.37$, positive-coupling rate 0.033; DS×Qwen: -1.61, rate 0.167); in the other nine conditions every repetition is positively coupled (rate 1.0), so the Bernoulli likelihood carries no information and the MLE is undefined. Where it is defined, the MLE has the same condition ordering as the pipeline proxy but takes negative values, violating the model's $\gamma > 0$ assumption. This confirms the Table 1 caveat that γ is not estimated from Bernoulli likelihood in the current pipeline: the seed-level coupling indicators are far too coarse to identify the logistic parameter, and the proposition's regularity conditions are correspondingly not satisfied at the self-evaluation conditions that dominate the low-γ regime.

3.2 Additional Assumptions Needed for a Product Bound

The observed negative γ–H association does not by itself imply a deterministic entropy law. We therefore state the missing assumption explicitly:

> **Entropy-floor assumption.** On the compact parameter region Θ and for a fixed TTRL update rule, normalized strategy entropy satisfies $H(\gamma) \geq h_{\min} > 0$.

Proposition 3 (Conditional product bound)**.** *Under the logistic observation model, regularity conditions, and entropy-floor assumption,*

$$\gamma H(\gamma)\mathrm{CV}(N) \geq \frac{2h_\min}{\sqrt{N}} > 0.$$

Proof. Proposition 1 gives

$$\gamma H\mathrm{CV} \geq \frac{H(\gamma)}{\sqrt{N s(\gamma)(1 - s(\gamma))}}.$$

Because $s(\gamma)(1 - s(\gamma)) \leq 1/4$ and $H(\gamma) \geq h_{\min}$, the right-hand side is at least $2h_{\min}/\sqrt{N}$. □

This proposition is conditional rather than universal. It fails to apply when observations are dependent, the estimator is biased/nonregular, γ approaches zero, or entropy has no positive floor. In particular, the first-order (Delta-method) approximation degrades for $\gamma \geq 1$, where $s(\gamma)$ approaches the 1 boundary and the 10/21 coverage failures concentrate (empirical-coverage paragraph above); the logit estimator is additionally biased as $\gamma \to 0$. The proposition therefore holds only in the regular-estimator regime; outside it, the bound is not a tight lower bound and must not be used for sample-size decisions. The empirical quantity $\sigma_w/(\|\mathbf{w}_0\|_2\sqrt{N})$ is retained only as a data-dependent heuristic for continuous weights; without a sampling theorem connecting σ_w to Fisher information, it is not labeled a tight lower bound.

4 Results: Empirical Validation

Statistical Note: Multiple Comparisons

We apply family-wise Bonferroni correction to control the family-wise error rate across multi-test families in this paper. For a family of k statistical tests answering a single empirical question,

Table 1: Theory-to-measurement map and assumption audit.

Symbol	Theoretical role	Empirical construction / limitation
γ	Scalar logistic parameter	Normalized distance between evaluator-conditioned and baseline strategy weights; not estimated from Bernoulli likelihood in the current pipeline
H	Entropy function with floor $h_{\min}$	Normalized entropy of final strategy weights; floor is assumed, not proved
CV(N)	Relative SD of regular $\hat{\gamma}_N$	Seed-level SD divided by seed-level mean; round dependence may violate independence
N	Independent sampling units	Number of seeds, not number of rounds or API calls
σ_w	Weight-coordinate dispersion	Empirical summary used in a heuristic continuous-weight expression
$\mathbf{w}_0$	Fixed initial weight vector	Protocol input; changes across protocols alter the estimand

the corrected threshold is $\alpha' = 0.05/k$. Significance stars below are reported against the corrected levels:

- *: $p < 0.05/k$
- **: $p < 0.01/k$
- ***: $p < 0.001/k$

The three multi-test families used in the empirical sections are:

- γ–H family ($k = 2$, $\alpha' = 0.025$): body 5-row $r = -0.9977$ ($p = 0.0001$); the abstract 11-LLM $r = -0.868$ is a source-study value that cannot be recomputed locally because the archive does not persist per-repetition H.
- Chatbot Arena correlation family ($k = 2$, $\alpha' = 0.025$): entropy–decisiveness $r(H, \text{dec}) = -0.895$; CV–decisiveness $r(\text{CV}, \text{dec}) = -0.075$.
- Convergence-dynamics family ($k = 2$, $\alpha' = 0.025$): half-life–α_{eval} $r = 0.76$; ΔH–α_{eval} $r = 0.95$.

Stars next to each p-value in the text below denote the corrected significance level; "NS" denotes non-significance at the corrected threshold.

4.1 Cross-Condition Validation (11 Conditions)

We validate across 11 evaluator–agent conditions spanning the source study's evaluator models (GPT-4o, DeepSeek-V3, Qwen-3.7, Claude-3.5) with three executor models and two experimental protocols; the archive reconstruction covers three of the four evaluator models (Claude-3.5 has no persisted row).

For the five reproduced rows, the Pearson correlation is $r = -0.9977$ ($p = 0.0001$), with leave-one-condition-out values ranging $[-0.9997, -0.9963]$; with $n = 5$, this estimate is highly sensitive to individual rows and is reported descriptively, and robust/permutation correlations require the missing condition-level data. This five-row set (DS Self-Eval, DS
*times*Qwen, Qwen
*times*DS, Ablation mid, Ablation max) is a different subset of the 11-condition survey than the companion paper's landscape subset ($r = -0.989$, $p = 0.0014$); both are reported with their

Table 2: Five condition rows as reported in the source study (CV(N=5) on the study's five-seed sets). The archive-based reconstruction of all eleven rows appears in Table 3 with per-repetition estimators; the two tables are not one-to-one (see mapping caveat below); this table therefore cannot support an all-11-condition audit.

Condition	Evaluator	γ	H	CV(N=5)	γHCV	Regime
DS Self-Eval	DeepSeek	0.033	0.999	2.42	0.080	Low-γ
DS × Qwen	DeepSeek	0.187	0.988	1.025	0.189	Intermediate
Qwen × DS	Qwen	0.766	0.931	0.157	0.113	High-γ
Ablation mid	Mixed	0.560	0.954	0.184	0.098	High-γ
Ablation max	Mixed	1.038	0.913	0.157	0.148	High-γ

The remaining six source-study conditions are represented by seven reconstruction rows in Table 3 (the mapping is not one-to-one; see the mapping caveat below).

Table 3: **Archive-based eleven-condition reconstruction.** γ = per-repetition mean; CV = sd/mean over the full archive; CV5 range = five-seed subsample distribution.

Condition	Archive	n	γ	CV	CV5 range
DS Self-Eval	mm_epc_multi_seed_ds_final.json	30	0.0332	5.477	[2.236, 2.236]
DS x Qwen (qwen eval)	mm_epc_multi_seed_final.json	30	0.1865	2.306	[0.587, 2.236]
Qwen x DS (official A)	official_A_qwen_eval_ds_exec.json	5	0.9866	0.241	[0.241, 0.241]
DS eval / Qwen exec (official B)	official_B_deepseek_eval_qwen_exec.json	5	0.8929	0.336	[0.336, 0.336]
GPT4o eval / DS exec len500 (official C)	official_C_gpt4o_eval_ds_exec_len500.json	5	0.7949	0.389	[0.389, 0.389]
GPT4o eval / Qwen exec (official D)	official_D_gpt4o_eval_qwen_exec.json	5	0.7238	0.357	[0.357, 0.357]
Ablation max	mm_epc_ablation_max.json	10	1.0384	0.366	[0.145, 0.511]
Ablation no-s0	mm_epc_ablation_no_s0.json	5	0.9790	0.400	[0.400, 0.400]
Qwen37	mm_epc_qwen37_final.json	8	1.0591	0.260	[0.143, 0.328]
GPT4o replication (OLD)	mm_epc_gpt4o_replication_OLD.json	8	1.1763	0.324	[0.208, 0.417]
GPT4o replication (June)	mm_epc_gpt4o_replication.json	8	0.0	0.0	[0.0, 0.0]
DS self-eval r30	mm_epc_ds_selfeval_r30.json	10	0.9360	0.196	[0.066, 0.286]

respective condition sets. The smallest observed product in the five-row table is 0.080 (DS Self-Eval: $0.033 \times 0.999 \times 2.42$); that fact does not identify the entropy floor or validate Proposition 3 because the empirical γ estimator does not follow the stipulated Bernoulli likelihood.

Archive-based eleven-condition reconstruction. Using the persisted per-repetition archives, we reconstruct all eleven conditions with a single estimator (per-repetition mean γ; CV over the full archive; five-seed subsample range): Table 3. The empty favorable region $\{\gamma < 0.2, \text{CV} < 0.3\}$ remains empty on the reconstruction: the two low-γ rows (DS Self-Eval 0.033, DS × Qwen 0.187) have full-archive CV ≥ 2.3, far above any reasonable CV cut.

Note. r30 protocol differs from multi_seed_ds_final; LOCO sensitivity: dropping this row changes CV 5.477→2.306, see Section 7.

Mapping caveat. Table 2 and Table 3 are not one-to-one: the source-study table reports CV(N=5) on specific five-seed sets and includes Ablation mid, whose archive is not persisted; the reconstruction reports per-repetition mean γ and full-archive CV with five-seed subsample ranges. Differences (Qwen×DS 0.766 vs. 0.9866; Ablation max CV 0.157 vs. 0.366; DS Self-Eval CV 2.42 vs. 5.477) follow from these estimator and seed-set differences. The repeated 2.236 ($= \sqrt{5}$) in the CV5 ranges is not a placeholder: for DS Self-Eval, 29 of 30 archived repetitions are zero-coupling, so any five-seed subset containing the single non-zero value has mean $x/5$ and sd $x/\sqrt{5}$, giving CV $= (x/\sqrt{5})/(x/5) = \sqrt{5}$ (using the sample standard deviation with Bessel-corrected denominator); all-zero subsets have undefined CV and are excluded. The DS Self-Eval row's low-γ/high-CV position rests on a single non-zero repetition (29/30 zero-coupling); per-seed γ values are provided in the evidence package, and this row should be treated as fragile when anchoring the frontier. The source-study CV(N=5)=2.42 for this row uses a different five-seed set and/or SD denominator than the reconstruction's 2.236.

Three robustness checks are computable from the persisted archives. (i) **Threshold robustness (full-archive CV estimator).** For every γ-cut in $[0.1, 0.5]$ and CV-cut in $[0.15, 1.0]$, the region $\{\gamma < \gamma_{\text{cut}},\ \text{CV} < \text{CV}_{\text{cut}}\}$ is empty for the archive reconstruction, because the two low-γ rows have full-archive CV ≥ 2.3. Subset-level emptiness is not claimed: five-seed subsample CVs can be as low as 0.587 for DS × Qwen (see (ii)), so a five-seed analysis could place that row inside a CV-cut of 0.6. (ii) **Seed-subset dependence of** CV(N=5)**.** Five-seed subsamples from the persisted 30-repetition archives yield CV in $[0.587, 2.236]$ (median 2.236) for DS × Qwen versus the reported 1.025, and the 252 five-seed subsets of Ablation max span $[0.145, 0.511]$ (median 0.355) versus 0.157; we therefore recommend reporting full-archive CV alongside CV(N=5). (iii) **Proxy–JSD alignment.** The empirical γ proxy correlates near-perfectly with the Jensen–Shannon distance between evaluator-conditioned and baseline strategy weights (Pearson $r = 0.969$, Spearman 0.998, $n = 152$; leave-one-source-out Pearson range $[0.968, 0.983]$), quantifying that the measured γ behaves as a distributional-distance proxy rather than the theorem's Bernoulli-likelihood estimator (per-seed γ–JSD pairs in the supplementary analysis output, N=152); the theory-to-estimator gap in Table 1 is therefore quantitative, not merely qualitative. (iv) **CV-estimator sensitivity.** Table 4 compares the full-archive CV (the estimator used for the reconstruction and the threshold-robustness checks) with the distribution of CV computed on five-seed subsamples of the same persisted archives. For the conditions with n=10–30 repetitions, the five-seed subsample median tracks the full-archive CV (Ablation max: 0.355 vs. 0.366; DS self-eval r30: 0.197 vs. 0.196; Qwen37: 0.250 vs. 0.260). The two n=30 self-evaluation archives are dominated by zero-coupling repetitions: five-seed subsamples either consist entirely of zero-coupling repetitions (for which CV is undefined) or collapse to the analytic $\sqrt{5}$ value derived in the mapping caveat above (DS Self-Eval: 2.236 for every non-degenerate subsample). We therefore use the full-archive CV for the empty-region analysis: it is the single estimator that is defined for all eleven conditions, maximizes sample efficiency, and is stable against the five-seed subsample distribution wherever that distribution is non-degenerate. The favorable regime $\{\gamma < 0.2,\ \text{CV} < 0.3\}$ remains empty under either estimator.

4.2 External Validation: Chatbot Arena (20 Models)

We analyze 55,000 publicly available human preference judgments from Chatbot Arena Chiang et al. (2024) across 20 models. The entropy–decisiveness correlation replicates: $r(H, \text{dec}) = -0.895$ ($p < 0.001^{**}$ under Bonferroni correction for the Chatbot Arena family with $k = 2$, $\alpha' = 0.025$). Models with clear pairwise dominance (GPT-4o, GPT-4-turbo) show lower outcome entropy than models with ambiguous comparisons.

Table 4: **CV-estimator sensitivity.** Full-archive CV vs. five-seed subsample CV (median with [min, max] range over non-degenerate five-seed subsamples). Script and output provided in supplementary material.

Condition	n	Full-arch. CV	CV5 median	CV5 range
DS Self-Eval	30	5.477	2.236	[2.236, 2.236]
DS x Qwen (qwen eval)	30	2.306	2.236	[0.587, 2.236]
Qwen x DS (official A)	5	0.241	0.241	[0.241, 0.241]
DS eval / Qwen exec (official B)	5	0.336	0.336	[0.336, 0.336]
GPT4o eval / DS exec (official C)	5	0.389	0.389	[0.389, 0.389]
GPT4o eval / Qwen exec (official D)	5	0.357	0.357	[0.357, 0.357]
Ablation max	10	0.366	0.355	[0.145, 0.511]
Ablation no-s0	5	0.400	0.400	[0.400, 0.400]
Qwen37	8	0.260	0.250	[0.143, 0.328]
GPT4o replication (OLD)	8	0.324	0.325	[0.208, 0.417]
DS self-eval r30	10	0.196	0.197	[0.066, 0.286]

Table 5: **The evaluation frontier.** The favorable regime $\{\gamma < 0.2, \mathrm{CV} < 0.3\}$ is empirically empty.

Regime	Example	γ	CV(N=5)
Low-γ, high-CV	DS Self-Eval	0.033	2.42
High-γ, low-CV	Ablation max	1.038	0.157
Low-γ, low-CV	*None*	—	—

The CV–decisiveness correlation is $r(\mathrm{CV}, \mathrm{dec}) = -0.075$ with 95

4.3 The Empirical Frontier

The frontier is bimodal: conditions cluster at either the low-γ, high-CV corner or the high-γ, low-CV corner. No condition occupies $\{\gamma < 0.2, \mathrm{CV} < 0.3\}$. At the reported point estimates this emptiness is threshold-robust within the five fully reported rows (both low-γ rows have CV(N=5) ≥ 1.025) and persists on the archive-based eleven-condition reconstruction (Table 3), where the two low-γ rows have full-archive CV ≥ 2.3.

5 Within-Condition Verification

5.1 Protocol Design

We provide a standardized protocol for within-condition verification of the bias-reliability tradeoff:

- Single LLM serves as both executor and evaluator
- Independent variable: evaluator feedback strength α_{eval} (recommended confirmatory levels $\{0, 0.5, 1.0\}$; the convergence-dynamics simulation below additionally probes $\{0, 0.4, 0.8, 1.5, 2.0\}$ to map the dose–response)
- Configuration: $N \geq 20$ seeds, $T \geq 30$ rounds, 6 tasks, 11 strategies
- Predictions: (P1) $\mathrm{corr}(\alpha_{\text{eval}}, \gamma_{\text{meas}}) > 0$; (P2) $\mathrm{corr}(\alpha_{\text{eval}}, H) < 0$; (P3) $\mathrm{corr}(\gamma_{\text{meas}}, \mathrm{CV}) < 0$

5.2 Convergence Dynamics

TTRL simulation (N=50 seeds, T=30 rounds) reveals three key findings:

Finding 1: Gradual, monotonic collapse. Strategy entropy $H(t)$ decreases smoothly from $H(0) \approx 0.9999$ to a steady-state value. No phase transition, no critical α_{eval}. Half-life: 14–19 rounds, largely independent of α_{eval} ($r = 0.76$, $p = 0.13$, NS under Bonferroni correction for the convergence-dynamics family with $k = 2$, $\alpha' = 0.025$).

Finding 2: Evaluator strength controls magnitude. Total entropy drop $\Delta H = H(0) - H(30)$ scales with α_{eval}: from $\Delta H = 0.0043$ at $\alpha_{\text{eval}} = 0$ to $\Delta H = 0.0128$ at $\alpha_{\text{eval}} = 2.0$ ($r = 0.95$, $p = 0.012^*$ under Bonferroni correction for the convergence-dynamics family with $k = 2$, $\alpha' = 0.025$).

Finding 3: Evaluator pressure increases heterogeneity. Across-seed variance in final entropy *increases* with α_{eval}: σ_{final} from 0.0022 to 0.0039. Stronger evaluator preferences produce greater heterogeneity in convergence outcomes.

Power analysis. The half-life of 14–19 rounds implies $T \geq 30$ for $> 85\%$ convergence. At T=30, the entropy difference between $\alpha_{\text{eval}} = 0$ and 1.0 has Cohen's $d \approx 1.2$, providing $> 80\%$ power at N=20 seeds. Recommended: 3–5 α_{eval} levels, 20–30 seeds per level, 30 rounds per seed. Cost formula: levels × seeds × rounds × tasks × calls-per-round; the recommended minimum (3 levels × 20 seeds × 30 rounds × 3 tasks × 1 call) is ≈ 5,400 API calls (≈ \$0.50 at the listed DeepSeek rates); a six-task battery at the same grid gives 10,800 calls, and the maximum ($5 \times 30 \times 30 \times 3$) is ≈ 13,500.

6 GPT-4o Version Drift

A single two-snapshot observation motivates longitudinal monitoring: GPT-4o coupling differs between two API snapshots one month apart ($\Delta\gamma = -1.176$; May 2026: $\gamma \approx 1.176$, N=8; June 2026: $\gamma = 0.000$, N=8, fully zero-coupling). The June snapshot's 100% zero-coupling needs seed-level corroboration; we treat this as a monitoring prompt, not an established drift law. Both snapshots used identical prompts, identical seeds, identical evaluation metrics; the only difference was the model snapshot. The claim that coupling measurements can invert within 4 weeks (magnitude $\Delta\gamma = 1.176$ at N=8) is refuted if a re-measurement in July 2026 under identical conditions yields γ within ± 0.2 of either the May or June 2026 value, or if a third non-zero snapshot in the same window produces $|1 - \gamma_{\text{July}}/\gamma_{\text{June}}| > 2$. Because the inversion went from strongly above-mean to strongly below-mean (and through zero, which is the canonical "uncoupled" regime), the actionable rule is: future studies must report model snapshot, version date, and repeat the measurement under at least $M \geq 2$ snapshots separated by ≥ 2 weeks; this rule is operationalized as a falsifiable design constraint (drift envelope ≤ 0.5 in γ across snapshots) and is refuted if any single-snapshot γ in a future study is reused more than 4 weeks after measurement without re-validation.

Implication (operationalized as a falsifiable design rule): coupling benchmarks must be versioned and re-validated, with re-validation required at intervals $\Delta t \leq 4$ weeks. The claim "a measurement valid today may be invalid next month" is operationalized as: a single-snapshot γ measurement loses operational validity if its magnitude drifts by $|\Delta\gamma| > 0.5$ within a 4-week window, with the specific evidentiary lower bound $|\Delta\gamma| \geq 1.176$ established by the May–June 2026 GPT-4o comparison (N=8 per snapshot). This is refuted if the July 2026 GPT-4o snapshot lies within $|\gamma - 0.000| \leq 0.2$ (i.e., June 2026 holds), or if $|\Delta\gamma| \leq 0.2$ between any two consecutive monthly snapshots over a 3-month follow-up.

7 Implications

7.1 For Experimental Design

1. **Use within-condition designs.** Between-condition comparisons confound evaluator identity with protocol parameters. Within-condition designs varying α_{eval} provide cleaner causal evidence.
2. **Use $N \geq 20$ seeds.** The decision-boundary simulation reported in a companion N-sensitivity study shows that at N=5 the false-positive probability for a decision-boundary conclusion is 13%; at N=30 it drops below 2%.
3. **Use $T \geq 30$ rounds.** Strategy distributions require $T \geq 30$ to approach steady state. Pilot studies with $T < 15$ will observe near-uniform weights regardless of evaluator strength.
4. **Report γ, H, and** CV **jointly.** The conditional product bound implies that reporting only one metric provides an incomplete picture.

Bound-guided sample allocation (decision case). The conditional precision floor $\text{CV}(N) \geq 1/(\gamma\sqrt{Ns(\gamma)(1-s(\gamma))})$ can be inverted to set confirmatory sample sizes before collection (script and output provided in supplementary material). Targeting CV=0.30 (the boundary of the empty favorable regime) yields the required seed counts: for the high-coupling conditions ($\gamma \in [0.72, 1.18]$) the bound demands $N \geq$ 45–96, for the intermediate conditions ($\gamma \approx 0.19$–0.99) $N \geq$ 53–1,289, and for the self-evaluation conditions ($\gamma \leq 0.19$) $N \geq$ 1,289–40,333. A naive "n=5 seeds suffice" rule is therefore rejected for all eleven conditions at this target; even the most favorable condition requires $N \geq 44$ when the bound applies. Because the coverage probe above shows that the logit estimator violates the bound's regularity conditions in 10/21 probed cells (small N or boundary $s(\gamma)$), the inverted N is a valid planning minimum only in the regular-estimator regime; in the failure cells the bound is non-tight in the opposite direction (empirical CV below the floor), so the inversion there would be unreliable and the planner should verify empirically before using it. This changes an allocation decision: an evaluator planner who uses the bound pre-registers N from the floor rather than from an experience rule, and for low-γ candidates either budgets the larger N or demotes them to exploratory status. Because Proposition 2 shows the floor can be non-tight (the proxy variance may far exceed the Bernoulli floor), the inverted N is a conservative reading of the required sample size: in the failure regime the inversion should be treated as the planning minimum and verified empirically before confirmatory use; in the regular-estimator regime it is conservative, in the failure regime the inversion is unreliable and should not be used for planning. This only strengthens the rejection of the n=5 heuristic, whose shortfall the bound already exposes. We confirm the allocation gain with a ranking simulation on the Chatbot Arena archive (20 models; script provided in supplementary material): recovering the full-sample top-5 model set, the n=100 per-model heuristic achieves 76.9% set-overlap, while the bound-guided allocation (per-model n^* from the inverted Bernoulli SE at SE=0.03) achieves 88.1% (+11.2 percentage points) at 2.72× the observations; exact top-5 order recovery rises from 0.7% to 2.7%. The bound therefore trades a predictable observation budget for materially more stable ranking decisions, whereas the heuristic under-samples the low-variance leaders and over-samples the high-variance tail.

7.2 For Evaluator Selection

The bimodal frontier (clusters at $\gamma \in [0, 0.2]$ with high CV, or $\gamma \in [0.7, 1.1]$ with low CV; the favorable regime $\{\gamma < 0.2, \text{CV} < 0.3\}$ is empty across 11 conditions) implies a binary choice:

- **Low-coupling evaluators** (e.g., self-evaluation): Unbiased but unreliable. Require $N \geq 30$ seeds for stable conclusions.

- **High-coupling evaluators** (e.g., cross-model): Biased but reliable. Produce stable measurements at $N \geq 5$ seeds, but the bias must be acknowledged.

There is no "best" evaluator Wolpert and Macready (1997) : the choice depends on whether the application prioritizes unbiasedness or reliability.

8 Discussion

What we know. A positive product floor follows under the explicit logistic model and entropy-floor assumption. The reproduced empirical grid shows a negative γ–H association and no point in the selected favorable region. It does not establish that a different evaluator, protocol, estimator, or threshold cannot populate that region. Chatbot Arena results are observational associations with different estimands.

What we don't know. Whether the conditional relation extends beyond TTRL to other adaptation mechanisms (e.g., prompt tuning, RLHF). Whether the favorable regime $\{\gamma < 0.2, \text{CV} < 0.3\}$ can be populated by multi-model ensembles or calibration interventions. A complementary perspective on what we don't know comes from recent bounded-memory testbeds for long-horizon agents Cheng et al. (2026): while our conditional product bound constrains the measurement side, the memory-contract view of Cheng et al. (2026) constrains the agent side, and their interaction (does an explicit memory contract help populate the favorable regime?) is unexplored. Self-evolving rubrics for skill use Zhu et al. (2026) and explicit agent abstention signals Luo et al. (2026) are two additional axes that could shift the empirical location of the bound and warrant investigation.

8.1 Limitations

The theoretical results assume binary observations and multiplicative reweighting. The empirical validation uses a limited set of LLM families. The GPT-4o version drift finding is based on two snapshots; continuous monitoring is needed to characterize drift dynamics. The reported CV(N=5) values are seed-subset dependent (five-seed subsample ranges: DS × Qwen $[0.587, 2.236]$; Ablation max $[0.145, 0.511]$), and the empirical γ proxy tracks JSD nearly perfectly (Spearman 0.998), so the theory-to-estimator gap is quantitative rather than merely qualitative.

Model-specific findings. The core results are established for the model families tested in this study. Whether the coupling effect generalizes to other model architectures, training recipes, or inference-time configurations is an open empirical question. Cross-model validation with larger sample sizes ($N \geq 30$ per condition) is needed to establish the generality of the findings.

Sample size and statistical power. Several experimental conditions use N=5 or N=8 seeds, which limits the precision of effect size estimates and the statistical power of hypothesis tests. The bootstrap confidence intervals we report quantify sampling uncertainty but cannot address systematic bias from seed selection or evaluator API version drift. Larger-scale replication with $N \geq 30$ per condition is recommended for confirmatory analysis.

External validity. All experiments use synthetic tasks with controlled communication protocols. Whether the findings generalize to real-world multi-agent deployments—where communication is asynchronous, tasks are open-ended, and agents have persistent memory beyond the current context—requires longitudinal field studies. The practical guidelines we provide should be interpreted as preliminary recommendations pending production validation.

Simulation-to-real gap. Where simulation results are reported, they use simplified dynamics models with fixed learning rates and strategy spaces. The simulation grid may not capture the full complexity of real LLM evaluator behavior, including context-length effects, token-level stochasticity, and API-level nondeterminism. Real API experiments are needed to validate simulation-based conclusions.

9 Conclusion

We provide a conditional precision and product bound for a stylized TTRL observation model and an explicit audit showing where the current empirical estimators depart from that model. The observed empty region and correlations remain descriptive until all condition rows, alternative metrics, threshold sensitivity, and held-out tests are available. This narrower contribution is preferable to a universal impossibility claim unsupported by the current observation model.

Broader Impact

The conditional product bound has implications for the safety of LLM-based evaluation systems, with quantitative magnitudes drawn from the 11-condition dataset: systems optimized for measurement reliability (high γ, e.g., Ablation max $\gamma = 1.038$, CV(N=5) = 0.157) produce stable measurements (low CV) but introduce measurable evaluator preference bias; systems optimized for unbiasedness (low γ, e.g., DS Self-Eval $\gamma = 0.033$, CV(N=5) = 2.42) produce unbiased estimates but with substantially higher measurement noise (CV inflated by 15.4× relative to Ablation max). The "may" framing of these safety claims is operationalized as: a high-coupling system introduces systematic bias if the induced ΔH (entropy drop from baseline) exceeds 0.05 at T=30; a low-coupling system produces unreliable measurements if CV ≥ 1.0 at N=5. In high-stakes applications where evaluation drives model selection, the tradeoff must be explicitly managed, with a falsifiable rule: model selection decisions that change the evaluator must be re-validated against the new evaluator's γ profile, and the rule is refuted if a re-evaluation under a new evaluator produces $|\Delta\text{accuracy}| \leq 0.01$ across a fixed benchmark (which would imply the bias has no measurable impact on selection).

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Reproducibility Statement

Frozen evidence. The numerical checks in this candidate are bound to a frozen evidence bundle with SHA-256 checksums. The audit distinguishes persisted API archives, deterministic reanalysis, and assumptions that remain outside the observed data.

Open metadata. This candidate does not assert that raw API logs, a public code repository, or a release license are already available. Those are submission-time decisions and must be bound to exact artifacts before the manuscript is uploaded.

Broader Impact and Ethical Considerations

This work studies how evaluator coupling affects measurement reliability in TTRL-style systems. The results are conditional on the stated model and should not be read as a certification that any evaluator is universally safe.

Evaluation risk. A model or API update can change the empirical coupling signal without an obvious change to the protocol. Practitioners should therefore bind evaluation results to model snapshots and re-run version-sensitive comparisons instead of treating a single-snapshot result as durable evidence.

Responsible reporting. We preserve negative results, estimator failures, and unresolved assumptions rather than converting them into stronger universal claims. The candidate package also distinguishes verified numerical checks from metadata and independent-review gates that remain open.